

Semantics of Comparative Constructions

Class 1: Gradable adjectives & degrees

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Part 1

Introductory remarks

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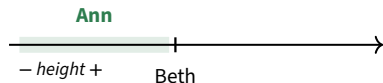
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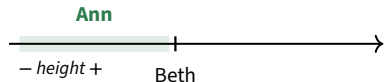
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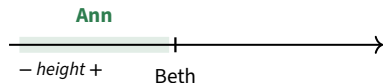
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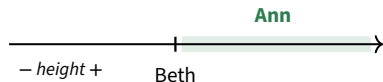
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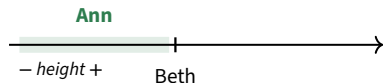
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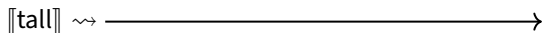
▷ **Our goal in this course:** develop a compositional semantic analysis of these constructions

Degree semantics

- This course is an introduction to analyses of comparatives based on the notion of **degrees**.

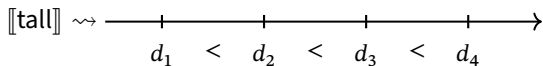
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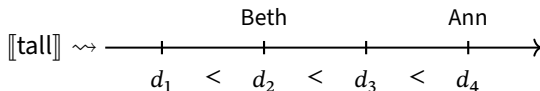
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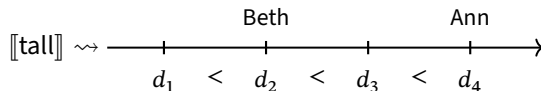
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- Degrees go beyond comparatives, but they are often the starting point into the phenomenon of gradability in natural language semantics.

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Limiting the scope

- Languages employ different strategies to express comparison, e.g.

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'He is taller than anyone.' (Warotamasikkkhadit 1972, via Stassen 2013)

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- However, given the amount of time we have we mostly focus on English comparatives:
 - most well-studied kinds of comparatives
 - complicated enough to keep us busy for the entirety of this introduction

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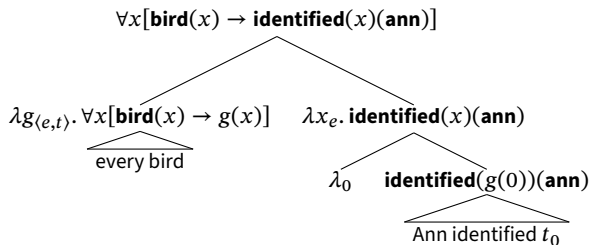
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- Specifically, I'll assume familiarity with:
 - (i) the semantics of quantificational DPs,
 - (ii) interpreting structures with instances of QR
- An example of what will be presupposed:

(9) a. Ann identified every bird.

b.



Today

- We said comparatives involve positioning two objects on a **scale**:
- (10)
- a. My dog is more **friendly** than yours.
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Side note. Of course, predicates other than adjectives can give rise to scalar meanings:

- (11)
- a. My grandma would fish more **often** than me.
 - b. My grandma would catch more **fish** than me.
 - c. My grandma would **fish** more than me.

The degree-based approach we discuss can be extended to these cases as well.

Part 2

Adjectives as predicates of individuals

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- We first consider **predicative** adjectives:

- (12)
- a. Ann is **vegan**.
 - b. The first dinosaurs were **bipedal**.
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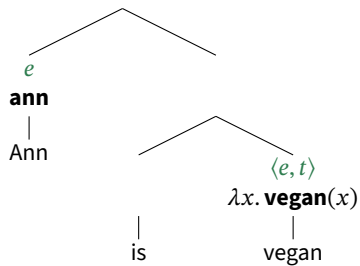
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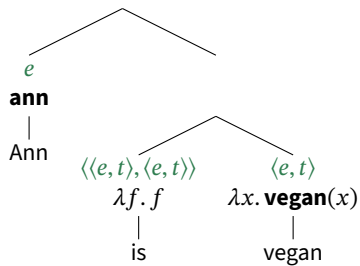
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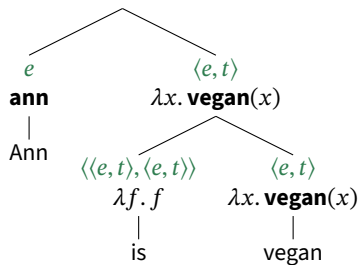
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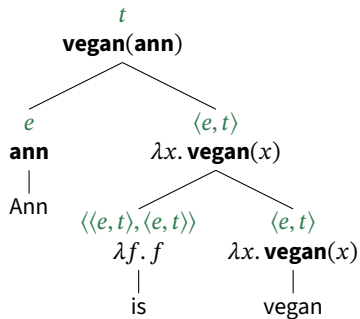
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What about attributive adjectives?

- The problem, however, is what we do about **attributive** adjectives:
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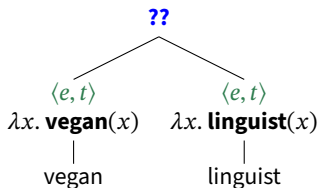
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- If adjectives are $\langle e, t \rangle$ functions, they can't compose with nouns via functional application:

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Interpreting attributive adjectives via Predicate Modification

- We thus introduced a new rule for interpreting complex syntactic nodes:

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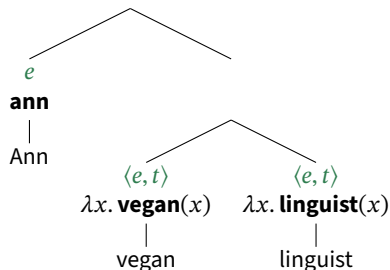
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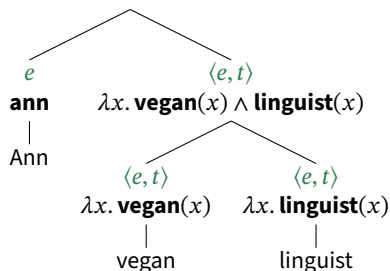
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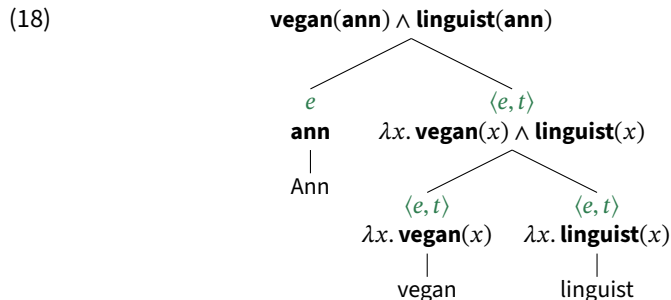
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On non-subjective adjectives

- The analysis of adjectives as $\langle e, t \rangle$ functions will only work for so-called **intersective** adjectives, i.e., adjectives that allow inferences such as the following:

(19) Ann is a vegan linguist.
Ann is a birder.
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- Standard analysis of these adjectives:** functions from (intensions of) NPs to NPs.

(for a great overview of different kinds of adjectives, see Ch. 2 of Morzycki 2016)

Part 3

Problems for the basic analysis

Adjectives and gradability

- Predicates associated with scales are also called **gradable**.
- Gradable predicates are those that can appear in comparatives and similar constructions:

- (23)
- a. The human femur is **longer** than the human tibia.
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- Seemingly non-gradable adjectives can often be coerced into gradable ones, however:

- (25)
- a. Ann is more **vegan** than Beth.
 - b. “the earliest dinosaurs were creatures that were completely **bipedal**.”
 - c. My shoes aren’t **waterproof** enough.

Gradable adjectives as $\langle e, t \rangle$ functions?

- Could we model gradable adjectives as $\langle e, t \rangle$ functions?

(26) a. $\llbracket \text{big} \rrbracket = \lambda x. \mathbf{big}(x)$ $::: \langle e, t \rangle$
b. $\llbracket \text{tall} \rrbracket = \lambda x. \mathbf{tall}(x)$ $::: \langle e, t \rangle$

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- This seems like a non-starter:
 - an $\langle e, t \rangle$ function is either true or false of an individual
 - we seem to need a notion of a predicate being true of something to **a certain degree**

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∴ Ann **drinks**.

- Substitution of identicals:
 - the second premise states *smoke* and *drinks* denote $\langle e, t \rangle$ function
 - thus, substituting one for the other doesn't change the truth of the first premise

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Everyone who **smokes drinks**, everyone who **drinks smokes**.

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(from 2017 lecture notes by Roger Schwarzschild)

Another problem: vagueness

- The $\langle e, t \rangle$ entry for *tall* below faces another issue:
 - it always maps an individual to true or false,
 - and thus assume there's a clear boundary between tall and non-tall individuals

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- We say *tall* in its positive form is **vague**:
 - some things are definitely a heap, others definitely aren't
 - but there are things in between & boundaries are fuzzy (=borderline cases)
- ⇒ this property is emblematic of many (but not all) gradable adjectives

Comparison classes

- Another properties of the positive form of adjectives is contextual variability:

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b. Paolo is tall **for a human**.

(for more on *for*-PPs see Kennedy 2007, Bale 2011, Bylinina 2012)

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- A possible paraphrase the meaning of positive adjective φ as:
 - from a comparison class, we compute a **standard of comparison**
 - $\llbracket \varphi \rrbracket$ is true of x when x exceeds the standard of comparison in φ -ness

Comparison classes & vagueness

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 If 150cm is tall for a 7yo, then so is 149cm.

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Comparison classes & vagueness

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(33) 150cm is tall for a 7yo.
If 150cm is tall for a 7yo, then so is 149cm.
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- **Kennedy 2007:** implicit comparison

(34) Compare to Beth, Ann is tall.

- * when Beth is 140cm and Ann is 160cm, the sentence is true
- * when Beth is 140cm and Ann is 141cm, ...

Part 4

Introducing degrees & scales

Adjectives as relations

- We now discuss a **degree-based** approach to the meaning of gradable adjectives.

(see Bartsch and Vennemann 1972, Seuren 1973, Cresswell 1976 Bierwisch 1989, Heim 1985, Stechow 1984, Kennedy 1999, among many others)

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(see Bartsch and Vennemann 1972, Seuren 1973, Cresswell 1976 Bierwisch 1989, Heim 1985, Stechow 1984, Kennedy 1999, among many others)

- In this approach, we will add **a degree argument** to gradable adjectives:

⇒ adjectives will be mapped to **relations between individuals and degrees**

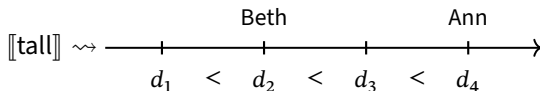
- (35)
- a. $x \text{ is } d\text{-tall} \approx x \text{ is tall to the degree } d$
 - b. $x \text{ is } d\text{-sharp} \approx x \text{ is sharp to the degree } d$
 - c. $x \text{ is } d\text{-colorful} \approx x \text{ is colorful to the degree } d$

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- **The idea in a nutshell (again):**

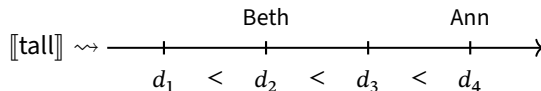
- a adjective φ is associated with an abstract measuring stick, i.e., a **scale**
- individuals are mapped to positions in this scale, i.e., to **degrees**
- degrees are ordered by a '**greater or equal than**' relation
- if x has more φ -ness than y , x is mapped to a greater degree than y is



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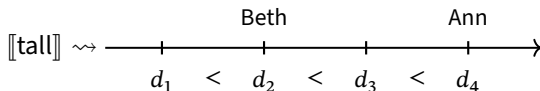
- This is all we need — and there are several ways of implementing this, e.g.

- degrees as equivalence classes of individuals (Cresswell 1976, Bale 2008, Schwarzschild 2020)
- degrees as kinds (of events) (Anderson and Morzycki 2015)
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⇒ We will make the simplest possible assumption and take degrees to be primitives.

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|---|------------------------------|
| • $d \geq_S d$ | reflexivity |
| • $d \geq_S d'$ or $d' \geq_S d$ | (strong) connectivity |
| • if $d \geq_S d'$ and $d' \geq_S d$, then $d = d'$ | antisymmetry |
| • if $d \geq_S d'$ and $d' \geq_S d''$, then $d \geq_S d''$ | transitivity |
| • if $d >_S d'$, then there's some $d^* \in S$ s.t. $d >_S d^*$ and $d^* >_S d'$ | density |

where $x > y := x \geq y \wedge x \neq y$

Introducing a new type

- In a basic extensional framework, we have two basic types e and t .
- We now introduce a basic type just for degree: d .

(36) **Semantic types**

- a. e , t and d are semantic types.
- b. If σ and τ are semantic types, $\langle \sigma, \tau \rangle$ is a semantic type.
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- c. Nothing else is a semantic type.

- We add to our model the set of all scales $\mathcal{S}$ and define the semantic domains as follows:

(37) **Semantic domains**

- a. $D_e := E$, the set of all individuals
- b. $D_t := \{0, 1\}$, the set of truth values
- c. $D_d := \{d \mid \text{for some scale } \langle S, \geq_S \rangle \in \mathcal{S}, d \in S\}$, the set of all degrees
- d. $D_{\langle \sigma, \tau \rangle} :=$ the set of all function from D_σ to D_τ

Part 5

Interpreting adjectives

Adjectives with measure phrases

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b. The fish I caught was **this** long.

(said while showing distance with hands)

Adjectives with measure phrases

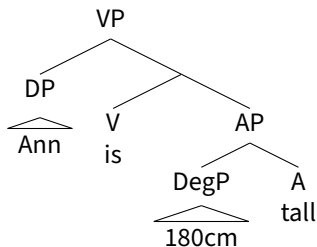
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- We assume that these sentences have the following underlying structure:

(39)



The type of adjectives

- Suppose we make the (unrealistic) assumption that *180cm* is the proper name of a degree:

$$(40) \quad \llbracket 180\text{cm} \rrbracket = \mathbf{180\text{cm}} \qquad \qquad \qquad :: e$$

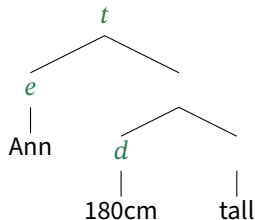
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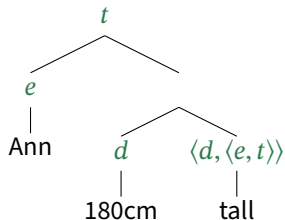
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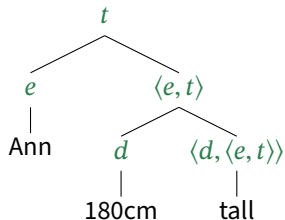
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‘Exact’ semantics for adjectives

- A simple analysis of *tall* would be to map an individual to its exact height:

$$(42) \quad \llbracket \text{tall}_{\text{exact}} \rrbracket = \lambda d. \lambda x. \mathbf{height}(x) = d \quad :: \langle d, \langle e, t \rangle \rangle$$

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 - a relation between the degree argument and the output of the measure function (in this case, **identity**)

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- Challenge:

$$(43) \quad \text{You have to be } \textit{this} \text{ tall to ride the roller coaster.} \quad (\textit{pointing to a line on the wall})$$

$\rightsquigarrow$ you have to be **at least** *this* tall

‘At least’ semantics for adjectives

- We thus change the relation between degrees in our entry:

$$(44) \quad \llbracket \text{tall} \rrbracket = \lambda d. \lambda x. \mathbf{tallness}(x) \geq_{\mathbf{tall}} d \quad :: \langle d, \langle e, t \rangle \rangle$$

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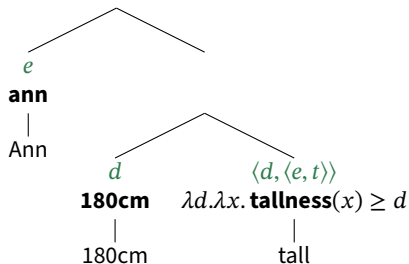
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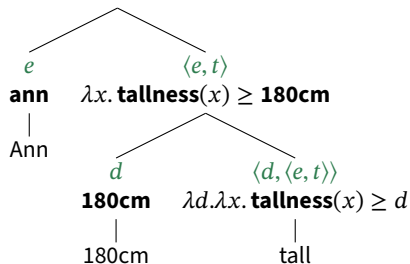
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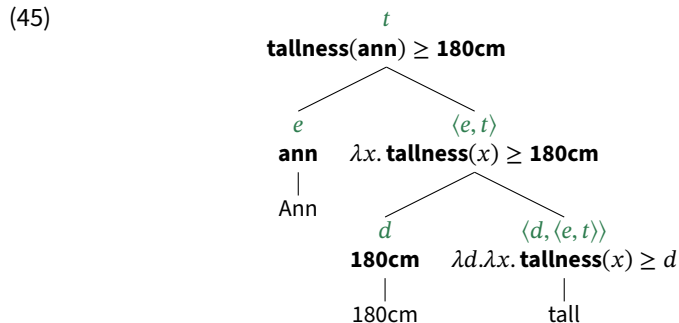
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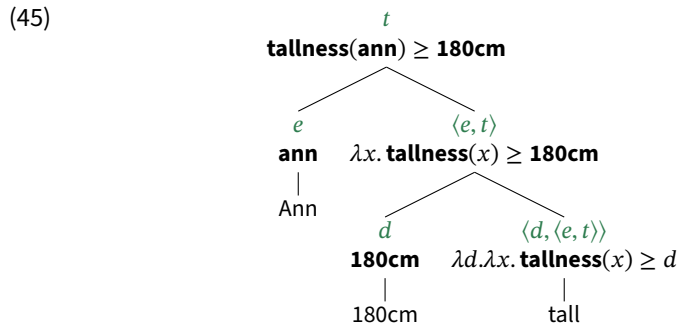
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↪ a so-called 'at least' semantics for the adjective

- Putting it all together:



- The meaning we get, then, is that *Ann is 180cm tall* is true if Ann's height is *at least* 180cm.
- We assume the 'exact' interpretation arises from pragmatic strengthening.

Part 6

On the semantics of the positive

The positive form of adjectives

- Adjectives their bare form–the **positive**–leads to vagueness and context variability:

- (46) a. My friend's 1yo son is tall.
b. My building is tall.

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- (46) a. My friend's 1yo son is tall.
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- Our meaning for adjectives say nothing about this aspect of bare adjectives:

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The positive form of adjectives

- Adjectives their bare form—the **positive**—leads to vagueness and context variability:

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- But in a way, that's by design! $\rightsquigarrow$ *vagueness disappears in (**some**) other forms of adjectives!*

- (48) Paolo is 1mm **taller** than my knees.
(49) I know **how tall** my friend Ann is: exactly 153cm.
(50) Ann is **150cm** tall.

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b. $\exists d. \llbracket \text{tall} \rrbracket(d)(\text{paolo}) \wedge d \text{ !> } \text{stnd}(\llbracket \text{tall} \rrbracket)(\llbracket 1\text{yo} \rrbracket)$

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- Entry for POS:

(53) $\llbracket \text{POS} \rrbracket = \lambda A_{\langle d, \langle e, t \rangle \rangle}. \lambda P_{\langle e, t \rangle}. \lambda x_e. \exists d. A(d)(x) \wedge d \text{ !> } \text{stnd}(A)(P)$

Part 7

Summing up

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 - * degrees form **scales**
 - * adjectives denote relations between degrees as individuals
- **Where we begin next time:** the treatment of vagueness in degree semantics.

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